

# Dipak Kumar Yadav

📍 Kathmandu, Nepal    📞 +977-9863052914    ✉ ydipak497@gmail.com    🌐 dippaakkWebsite  
🔗 dippaakk    📄 dipak497

## RESEARCH INTERESTS

---

My research in particle physics phenomenology is grounded in Quantum Field Theory, focusing on precision tests of the Standard Model (particularly the Higgs boson and electroweak symmetry breaking) and searches for new physics (BSM). I develop and apply deep learning and high-performance computing techniques to extract subtle signals from large-scale collider data, bridging the gap between theoretical models and experimental observation

## EDUCATION

---

**B.Sc. Physics (Four-Year Programme)**, St. Xavier's College, Maitighar, Kathmandu, Nepal 2020–2025  
**Percentage: 71% (First Division, approx. GPA ~.....).**

Selected Coursework: Quantum Mechanics; Solid State & Nuclear Physics; Elementary Particle Physics; Classical Mechanics; Electrodynamics; Computational Physics; Differential Equations; Linear Algebra; Probability & Inference.

**High School in Science**, Lord Buddha Secondary School, Biratnagar, Nepal 2018–2020  
Completed 10+2 science curriculum with excellent academic record. **GPA = 3.37/4,**

## RESEARCH EXPERIENCE

---

**Identifying Higgs Boson Events in  $\sqrt{s} = 13$  TeV ATLAS Data Using Deep Neural Networks** 2024–2025

Undergraduate Thesis Project, Department of Physics, St. Xavier's College, Tribhuvan University, Nepal  
Supervisor: Dr. Vinaya Kumar Jha

- **Engineered** and trained a deep neural network (TensorFlow/Keras) to identify Higgs boson signals in simulated ATLAS open source data, achieving a state-of-the-art classification performance of AUC = 0.94.
- **Designed** a data analysis pipeline using ROOT to process Monte Carlo events in the  $H \rightarrow ZZ \rightarrow 4\ell$  decay channel, extracting key kinematic features to optimize model performance.
- **Demonstrated** discovery-level sensitivity by achieving a statistical significance of  $5.37\sigma$  in a high-purity signal region, and validated the model's efficacy by benchmarking against traditional analytic methods.

**Deep and Shallow Neural Networks in Particle Physics Experiments** Jul–Aug 2025  
Summer Research Internship, IFJ PAN Particle Physics Summer Student Programme (PPSS 2025), Kraków, Poland  
Supervisors: Dr. hab. Tomasz Wąchała Dr. Karol Adamczyk

- **Benchmarked** deep vs. shallow neural networks for signal classification across diverse HEP datasets (ATLAS Higgs, MiniBooNE neutrino, Belle II continuum), demonstrating that deep architectures achieved superior performance (AUC 0.88 vs. 0.83) on complex, high-dimensional data.
- **Achieved** discovery-level sensitivity on the 11-million-event Higgs benchmark dataset, obtaining a  $5.36\sigma$  statistical significance by optimizing the deep learning model for a high-purity signal region.
- **Implemented** a GPU-accelerated training workflow on the Cyfronet Ares supercomputer, reducing

model training time by over 6x (from 150s/epoch on CPU to 23s/epoch on GPU) to enable large-scale analysis.

- **Completed** a competitive 160-hour programme combining supervised research with intensive lectures and hands-on training in core HEP tools (ROOT, Hypatia).

## PRESENTATIONS & POSTERS

---

**Poster: “Identifying Higgs Boson Events in  $\sqrt{s} = 13$  TeV ATLAS Data Using Deep Neural Networks”**

Presented at the Pentecost Spring School on Quantum Science 2025 (**PenteQost-25**), University of Siegen, Germany Jun 6–10 2025

**Talk: “Deep and Shallow Neural Networks in Particle Physics Experiments”**

Co-presented with Kacper Kopeć at the **PPSS 2025 Mini-Conference**, IFJ PAN, Kraków, Poland Aug 1 2025

Slides available at [ppss.ifj.edu.pl/materials\\_2025/12\\_PPSS\\_Kopec\\_Yadav.pdf](https://ppss.ifj.edu.pl/materials_2025/12_PPSS_Kopec_Yadav.pdf).

## WORKSHOPS, SUMMER SCHOOLS & COMPETITIONS

---

**Pentecost Spring School on Quantum Science 2025 (PenteQost-25)**

June 6–10, 2025

University of Siegen, Germany

- Participated in an interdisciplinary spring school covering topics such as Optical Clocks & Precision Spectroscopy, Quantum Foundations, Open Quantum Systems, Quantum Information, Statistical Inference, Integrable Quantum Systems, and Quantum Computational Complexity and lab tour of Germany’s first Quantum computers. .

**Youth in High-Dimensions: Recent Progress in Machine Learning and High-Dimensional Statistics** July 3–10, 2025

International Centre for Theoretical Physics (ICTP), Trieste, Italy (*online via Zoom*)

- Attended lectures on recent advances in machine learning, high-dimensional inference, and philosophical perspectives on generative AI.

**PLANCKS 2024 Finals (International Association of Physics Students)**

May 23–27, 2024

Trinity College Dublin, Ireland

- Represented Nepal in the international theoretical physics competition; competed in advanced problems across quantum mechanics, relativity, classical mechanics, and statistical physics.

**Summer School in Theoretical (Astro)Physics (SSTP-2024)**

2024

IUCAA & St. Xavier’s College, Ahmedabad, India

- Studied relativity and applied machine learning techniques to analyze gravitational-wave data from GWOSC.

**BCVSPIN-2024: Masterclass in Particle Physics (ML Workshop)**

Dec 4–6, 2024

Central Department of Physics, Tribhuvan University, Nepal

- Introduced to particle physics & HEP experiments; hands-on ML with ATLAS data; group presentation on Higgs signal–background classification.

**Holographic Himalaya SS2 (ICTP PWF Nepal)**

2023

Tribhuvan University, Kathmandu, Nepal

- Report writing on *Foundations of Quantum Mechanics* and a short research internship under the ICTP Physics Without Frontiers (PWF) initiative.

**Additional Workshops:** Completed workshops on *Introduction to Quantum Computing* (Qiskit) and *Quantum ESPRESSO Basics* for ab-initio simulations (15 hours / 1 credit).

## HONORS, AWARDS & GRANTS

---

**Fr. Bertrand Saubolle, S.J. Memorial Award (Outstanding Student of the Batch)** 2025

A prestigious college-wide award recognizing a graduating student for all-around excellence in academic competitions, leadership, and contributions to international research.

**Winner, “10 Years of PLANCKS” Challenge** 2024

Awarded for top performance in theoretical-physics problem solving at the PLANCKS 2024 international finals in Dublin.

**PLGrid Computational Grant** 2025

Granted access to the Ares HPC cluster at the Cyfronet Centre, enabling large-scale deep-learning research for the PPSS-25 programme.

**Research & Travel Grants** 2025

Awarded competitive funding for the PenteQost Spring School (€700) and the IFJ PAN Summer Student Programme (NPR 30,000).

**Undergraduate Merit Scholarship** 2020–2024

A four-year academic scholarship from St. Xavier’s College awarded for outstanding and sustained academic performance in physics.

## COURSES & CERTIFICATIONS

---

**Diploma in Quantum Computing and Programming** — QWorld & Qiskit 2024

Instructor: Dr. Jibran Rashid Diploma No. Qbronze115-60

**Particle Physics: An Introduction** — University of Geneva, Coursera 2024

**Machine Learning Introduction for Everyone** — IBM, Coursera 2024

**Programming for Everybody & Python Data Structures** — University of Michigan, Coursera 2023

## TEACHING & WORK EXPERIENCE

---

**Teacher (Science & Mathematics)**, Mrigashira World School, Kathmandu, Nepal 2023–2024

- Designed an engaging curriculum for high-school (secondary) science and mathematics, using interactive discussions and hands-on activities to foster critical thinking and deepen conceptual understanding.

**Private Tutor (Physics & Mathematics)**, Independent 2020–2023

- Mentored high-school students in advanced physics and mathematics, focusing on developing critical-reasoning and problem-solving strategies to improve exam performance.

## LEADERSHIP, EXTRACURRICULAR ACTIVITIES & OUTREACH

---

**Executive Representative (2022–2023) & Treasurer (2023–2024)**, St. Xavier’s Physics Council Nepal

- Organised and coordinated key academic and outreach initiatives to foster student engagement, including leading events such as *Yuri’s Night*, *Pi Day*, and quantum-computing workshops.

**Rural Immersion Camp** (During 2<sup>nd</sup> & 3<sup>rd</sup> Year Undergraduate), Various remote villages in Nepal

- Engaged in immersive fieldwork in remote rural communities to analyze grassroots socio-economic challenges and contribute to local outreach and educational support.

## SKILLS

---

**Programming & Scripting:** Python, C, C++, HTML, MATLAB,  $\text{\LaTeX}$

**Data Analysis & Machine-Learning Frameworks:** ROOT, HYPATIA, TensorFlow, Keras, scikit-learn

**Systems & Development:** Linux, Git/GitHub, Docker, SLURM